

Lab-based activity 5.1: Laboratory works

Objectives**:** observe and identify three cell types

Materials

Microscopes, cardiac, skeletal, and smooth muscles specimens, and Knives.

Methods

- Prepare specimen for all the three muscle types
- Set a wet mounting of each specimen taken from each muscle cell
- Observe the specimen samples under the microscope
- Write a lab report on the protocol.

Assessments

What are the differences among the three types of muscles?

Draw a diagram of each type of muscle

Explain the relationships between the structures and functions of muscles.

5.1.2. Mechanism of actions of skeletal muscles

What are mechanisms of action?

Mechanisms of action of muscles are the process of activations of muscle cells to generate tension or contract for movements.

Skeletal muscles are bundles of muscle fibers with single large cells formed by the fusion of many individual cells during development. Most of the cytoplasm consists of myofibrils of cylindrical bundles of two types of filaments of thick filaments of myosin and thin filaments of actin where each is each organized as a chain of contractile unit called sarcomeres.

Actin and myosin are both proteins found in every type of muscle tissue where the thick myosin filaments and thin actin filaments work together to generate muscle contractions and movement.

Myosin is a type of molecular motor that converts chemical energy released from ATP into mechanical energy used to pull the actin filaments along causing muscle fibers to contract and generate.

Actin filaments, together with myosin are responsible for many types of cell movements. The most noticeable form of movement is muscle contraction, which has provided the model for understanding actin-myosin interactions and the motor activity of myosin molecules.